day 14 | 250929 M

hitherto

We discussed old homework assignments, and you will turn them in to me today. The homework assignments are exercises 3.8, 5.1, 5.2, and 5.3.

herein

We will discuss some other integration methods. We will talk though some quadritures (which is an old word for numerical integration). I have some defined below, but we will also use a function that is provided to us by the author, namely `gaussxy.py` which has a couple of methods within it that we will use. It is important to note that you need to have this python script put in the same folder that you are going to call it from, so whether that is here in `develop` for today, or a copy in `deliver` for any homework, you'll need it in both places. Also, as I mentioned in the email I sent you, this particular bit of script has been updated by the author, so the version that you have in your `data` folder will only work if you look at it and make it work for you.

We also looked at integrals over infinite ranges, and we used a change in variables to handle those situations. In most cases you can make the following substitution:

$$z = \frac{x}{1+x} \quad \longleftrightarrow \quad x = \frac{z}{1-z}$$

There are other variations on this that you should be aware of like

$$z = \frac{x-a}{1+x-a}$$

but in either case

$$dx = \frac{dz}{(1-z)^2}$$

Now you can integrate either from 0 -> $\inf$ or a -> $\inf$. To do the other side like from $-\inf$ to 0 or to a, then make z -> -z up above.

hence

We will practice this and work exercises 5.9 and 5.12 next.

In [2]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
```

In [7]:
```
def GaussLagQuad8(function):
    x=np.asarray([1.7027963230510100e-1, 9.0370177679937991e-1,\
       2.2510866298661307,    4.2667001702876588,\
       7.0459054023934657,    1.0758516010180995e+1,\
       1.5740678641278005e+1, 2.2863131736889264e+1])
    w=np.asarray([3.6918858934163753e-1, 4.1878678081434296e-1,\
       1.7579498663717181e-1, 3.3343492261215652e-2,\
       2.7945362352256725e-3, 9.0765087733582131e-5,\
       8.4857467162725315e-7, 1.0480011748715104e-9])
    integral = np.sum(w*function(x))
    return(integral)

def gaussHermQuad8(function):
    x = np.asarray([-0.38118699,-1.157193712,-1.981656757,-2.93063742,0.3811
    w = np.asarray([0.661147013,0.207802326,0.017077983,0.000199604,0.661147
    integral = np.sum(w*function(x))
    return(integral)

def gaussLegQuad8(function):
    x = np.asarray([0.183434643,0.52553241,0.796666477,0.960289857])
    # x = np.asarray([-0.183434643,-0.52553241,-0.796666477,-0.960289857,0.1
    w = np.asarray([0.362683783,0.313706646,0.222381035,0.101228536])
    # w = np.asarray([0.362683783,0.313706646,0.222381035,0.101228536,0.3626
    integral = np.sum(w*function(x))
    return(integral)
```

In [4]:
```
def f(x):
    return(x**4 - 2*x + 1)
```

In [5]:
```
gaussLegQuad8(f)
```

Out[5]: 2.399999999992067

In [6]:
```
def g(z):
    return(np.exp(-z**2/(1-z)**2)/(1-z)**2)
```

In [8]:
```
gaussLegQuad8(g)
```

Out[8]: 0.925771851897418

In [9]:
```
.5*np.sqrt(np.pi)
```

Out[9]: 0.8862269254527579

In [10]:
```
def trap(func, a, b, steps):
    dx = (b-a)/steps
    x = np.linspace(a, b, steps+1)
    y = func(x)
    s = dx/2*(func(a)+func(b))
```

```python
    s = s + np.sum(y[1:steps]*dx)
    return(s)
```

In [13]:
```python
trap(g, 0, .99999, 100)
```

Out[13]:
```
0.8862102591194666
```

In [14]:
```python
from gaussxw import gaussxwab
```

In [56]:
```python
N = 20

x, w = gaussxwab(N, 0, 1)
s = 0.0
for k in range(N):
    s += w[k]*g(x[k])
```

In [57]:
```python
s
```

Out[57]:
```
0.8862261838125776
```

In [ ]:
```python

```